

Quiz 8

● Graded

Student



Total Points

17 / 30 pts

Question 1

(no title)

8 / 11 pts

+ 11 pts Correct



+ 10 pts Correct with minor error

✓ + 6 pts Useful observations made but either there is something incorrect or not the key observation

+ 2 pts incorrect attempt made

+ 0 pts No attempt is made

🗨 + 2 pts orthogonal means that $A^T = A^{-1}$ so $A^T A = I$ was the observation I was looking for

Question 2

(no title)

5 / 8 pts

+ 8 pts Correct

✓ + 5 pts Correct answer but poor justification

+ 4 pts incorrect answer but correct justification

+ 2 pts incorrect answer but justification shows some understanding

+ 0 pts Incorrect

Question 3

(no title)

4 / 11 pts

+ 4 pts First part correct

+ 3.5 pts First part correct but missing details

+ 2 pts Made partial progress in first part

✓ + 0 pts left blank or no progress made

+ 7 pts Second part correct

+ 6 pts Second part almost completely correct

✓ + 4 pts Made partial progress in second part

Math 54 Quiz 8

Name: 

ID: _____

Problem 1

Find an expression for a solution to the least squares problem $Ax \approx b$ when the matrix A is orthogonal, i.e. when the columns are orthonormal. Explain briefly why this case is particularly nice.

When A has the properties of orthogonality ^{orthonormality}

$$\rightarrow \hat{x} = [A^T A \mid A^T x] = x - [ERO \mid A^T A \mid A^T x] \sim (A^T A)^{-1} (A^T x)$$

This is extremely nice for computation of $n \times n \vee m \times n$ when we have linear independence, yielding $O(n^3)$ computation time.

$$\beta_j := \text{output of } x_j \text{ with } \sum_{j+1}^n \text{ with all } x_{j-1} \text{ fixed}$$

Problem 2

(T/F) In multi-variable linear regression, the coefficient β_j measure the change in the output in accordance with a one-unit increase in the parameter x_j , while holding the other parameters fixed. (Justify briefly)

True, because we are trying to approx. the actual data with $y = X^T X \beta / X^T y$ of actual $y = \text{observed } y$

Meaning $\forall x_{j-1}$ at current step x_j has to be fixed by induction hypothesis

Problem 3

Fill in the gaps in the following proof of the Cauchy-Schwarz inequality for $\mathbb{R}^n$. Recall that the Cauchy-Schwarz inequality says that for $u, v \in \mathbb{R}^n$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Proof. First, suppose that one (or both) of u or v is the zero vector. Then, _____. Now we assume that neither of u or v is the zero vector, so $\|u\|, \|v\| \neq 0$. Hence, it suffices to prove that

$$\left| \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle \right| \leq 1.$$

Observe that the left hand side is just the length of the projection $\text{proj}_{\frac{v}{\|v\|}}(\frac{u}{\|u\|}) = \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle \frac{v}{\|v\|}$. However the length of this projection is ≤ 1 because _____. (Please don't actually use the underlines to fill in your answer, they are far too small.)

1) Then, assume additivity and homogeneity are closed $\in \mathbb{R}^n$
for u, v

2) the inner product of $\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \rangle$ for u, v will always
be a smaller multiple of $\|u\| \cdot \|v\|$ in radical form
or $2^k \vee 2^{k+1}$